Reproducing *Physics of Language Models: Part 1, Context-Free Grammar*

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Abstract

We present a partial reproduction of Allen-Zhu and Li [1] (*Physics of Language Models: Part 1, Context-Free Grammar*), which studies how transformer language models learn the latent hierarchical structure of context-free grammars (CFGs). We re-implement the training and evaluation pipelines, train a GPT-2 model with absolute positional embeddings on the `cfg3b` synthetic grammar for 6,500 steps, and evaluate it through completion accuracy, KL divergence, linear probing of hidden states, and attention-pattern analysis. Our results largely confirm the qualitative findings of the original paper, while quantitative gaps—particularly in KL divergence—are attributable to the reduced training budget.

1 Introduction

A central question in the study of large language models is whether next-token prediction alone is sufficient for a model to discover the latent structure of the language it is trained on. Allen-Zhu and Li [1] investigate this in a controlled setting: synthetic languages generated by context-free grammars (CFGs), where ground-truth parse trees are available by construction. This allows precise, quantitative questions to be asked about the model's internal representations.

We reproduce their experimental setup on the `cfg3b` grammar using a standard GPT-2 architecture. Our goal is to verify whether the original conclusions hold under a reduced training budget (6,500 vs. the paper's 100,000 iterations), and to understand the gap between our results and the paper's targets.

2 Background

2.1 Context-Free Grammars as Synthetic Languages

A context-free grammar $G = (V_N, V_T, R, S)$ defines a language over terminal symbols V_T via rewriting rules in R. Each string $x = (x_1, \ldots, x_n)$ is derived from the start symbol S through L levels of non-terminal expansions. At each level ℓ, every terminal position i has a unique ancestor non-terminal symbol $s_\ell(i)$ and ancestor index $p_\ell(i)$. A *boundary* at level ℓ and position i is defined as:

$$b_\ell(i) = \mathbf{1}[p_\ell(i) \neq p_\ell(i+1)],$$

i.e., position i is the last terminal of a constituent at level ℓ.

The grammar `cfg3b` contains 32 production rules over 16 non-terminal symbols and 3 terminal symbols, with a hierarchy approximately 7 levels deep and branching factors of 2 or 3.

2.2 Probing Hidden States

Following Allen-Zhu and Li [1], we use *multi-head linear probing* to test whether the hidden state $h_i \in \mathbb{R}^d$ at position i encodes structural information. A probe $f_\theta(h_i)$ is trained to predict the ancestor symbol $s_\ell(i)$.

Two probe variants are used:

Full probe (R4). Unrestricted attention; accuracy evaluated over all positions.

Diagonal probe (R5). Attention restricted to a band of width $2\delta + 1$ around the diagonal; accuracy evaluated at boundary positions only.

3 Methods

3.1 Model and Training

We train a GPT-2 model [2] with learned absolute positional embeddings (`gpt_abs`): 12 layers, 12 heads, embedding dimension $d = 768$, sequence length 512, vocabulary size 5.

Training uses AdamW (β_1=0.9, β_2=0.98, weight decay 0.1, learning rate 3×10^{-4}) with a linear decay schedule over N=6,500 steps. We use gradient accumulation over 8 micro-batches of size 12 (effective batch size 96). Strings are sampled on-the-fly from `cfg3b`.

3.2 Evaluation

We evaluate the model on four tasks:

Completion accuracy (R1). The model generates a completion autoregressively from a prefix of length 0 or 50. A completion is correct if the full string is a valid string of the grammar. Reported under both multinomial sampling and greedy decoding.

KL divergence (R3). We compute $\mathrm{KL}(P_{\mathrm{true}} \| P_{\mathrm{model}})$ averaged over positions across 200 sampled strings.

Probing (R4–R5). For each non-terminal level ℓ, we train a multi-head linear probe for 5,000 iterations on frozen hidden states.

Attention analysis (R6–R9). We measure how attention is distributed relative to token distance and constituent boundaries.

4 Results

4.1 Completion Accuracy and KL Divergence

Table 1 summarises our quantitative results on `cfg3b` after 6,500 training steps.

Table 1: Quantitative results for `gpt_abs` on `cfg3b` (6,500 steps).

Metric	Value
Completion acc. (multinomial, no prefix)	5.50%
Completion acc. (multinomial, prefix = 50)	8.50%
Completion acc. (greedy, no prefix)	0.00%
Completion acc. (greedy, prefix = 50)	16.50%
KL divergence (nats/token)	0.01234
Paper target (GPT_rot, cfg3b)	≈0.00008

Completion accuracy is low in absolute terms, which is expected given our reduced training budget of 6,500 steps compared to the 100,000 steps used in the original paper. The greedy decoder with no prefix collapses to a repetitive sequence (0.00%), a known failure mode of greedy decoding on language models that have not converged. With a 50-token prefix, greedy decoding reaches 16.50%, showing that the model benefits substantially from conditioning on a partial context.

The KL divergence of 0.01234 nats/token is two orders of magnitude above the paper's target of ≈0.00008 for a fully trained model on the same grammar. This gap is consistent with early-stage training: the model has learned a rough approximation of the distribution but has not yet converged to the true grammar distribution.

4.2 Probing (Results 4–5)

We train multi-head linear probes on frozen hidden states for each non-terminal level $\ell \in \{2, \dots, 6\}$ of `cfg3b`. Table 2 reports full-probe accuracy (R4, evaluated over all positions) and diagonal-probe accuracy (R5, evaluated only at boundary positions with $\delta = 0$).

Table 2: Probing accuracy for `gpt_abs` on `cfg3b` (6,500 steps). Paper targets are for a fully-trained GPT_rot on cfg3f.

Level	Full R4	Diagonal R5
NT2	53.3%	87.1%
NT3	59.1%	66.3%
NT4	78.0%	89.5%
NT5	86.1%	99.9%
NT6	87.1%	91.9%
Paper target	≈94–100%	≈97–100%

Full-probe accuracy (R4) improves at deeper levels (closer to the terminal layer), reaching 87.1% at NT6. This is consistent with lower-level constituents being smaller and more locally decodable. The shallower levels—NT2 and NT3—show weaker accuracy (53.3% and 59.1%), which we attribute to insufficient training: at 6,500 steps the model has not yet fully encoded higher-level constituency.

The diagonal probe (R5) consistently and substantially outperforms the full probe trained for the same number of iterations. The gap is striking: at NT5, R5 reaches 99.9% versus R4's 86.1%; at NT2, R5 reaches 87.1% while R4 only reaches 53.3%. This performance advantage, achieved with identical compute, directly supports the original paper's central claim: *the hidden states at NT-boundary positions already encode the full ancestry information of the constituent they close*. Because the diagonal probe is restricted to a narrow window around position i, its high accuracy means the relevant information is locally available at boundary positions without needing global context. At non-boundary positions, the hidden state is consistent with multiple possible subtrees, making classification information-theoretically impossible—a result formalised in the original paper.

4.3 Attention Analysis (Results 6–9)

Result 6 — Distance-based attention.. Averaging attention weights $\bar{A}_p$ over all pairs (i, j) with $j - i = p$, across all layers and heads, we observe a strong recency bias. The distribution peaks at $p = 1$ ($\bar{A} = 0.0946$) and $p = 2$ ($\bar{A} = 0.0927$), decaying rapidly thereafter (e.g. $\bar{A} = 0.0097$ at $p = 15$). A uniform distribution over 100 positions would give $\bar{A} \approx 0.01$, so the model attends roughly $9\times$ more to the immediately preceding token than baseline. This distance bias is subtracted to isolate structure-sensitive attention.

Result 7 — Attention to NT-boundaries.. After removing the distance bias, the residual $B = A - \bar{A}$ is measured for key positions at offset δ from a constituent boundary. Table 3 shows a clear peak at $\delta = 0$ across all levels: the model disproportionately attends to exact boundary positions. The effect is strongest at NT2 ($B = +0.0085$) where boundaries are rare and summarise large subtrees, decreasing toward NT6 ($B = +0.0019$) where boundaries are frequent and subtrees are small.

Table 3: Result 7 — Residual attention B at offset δ from NT-boundary.

	δ=−2	δ=−1	δ=0	δ=+1	δ=+2
NT2	−.0008	−.0018	+**.0085**	−.0024	−.0013
NT3	−.0006	−.0017	+**.0065**	−.0019	−.0012
NT4	−.0005	−.0016	+**.0064**	−.0018	−.0009
NT5	−.0003	−.0015	+**.0047**	−.0006	−.0010
NT6	+.0006	−.0016	+**.0019**	−.0008	−.0002

Result 8 — Boundary-to-boundary attention.. Restricting both query and key positions to NT-boundaries at the same level, mean residual B is positive at all levels: NT2 +0.0020, NT3 +0.0056, NT4 +0.0080, NT5 +0.0059, NT6 +0.0020. The peak at NT4 coincides with the level where boundary density and subtree size are best balanced, providing the most informative pairings.

Result 9 — Adjacent NT-end attention.. Measuring B as a function of ancestor distance r between same-level boundary positions, the signal is consistently maximised at $r = 1$ and decreases with r. For example, at NT5: $r = 1$: +0.0315, $r = 2$: +0.0211, $r = 3$: +0.0038. At NT2 the signal is near zero for $r \geq 2$ due to the small number of high-level constituents per string. This monotone decay is the signature of a DP-like recurrence: the model retrieves structural information primarily from the immediately preceding constituent boundary, mirroring CYK-style parsing.

5 Extension: Present vs. Used Information

5.1 Motivation and Target Level

The probing experiments (R4–R5) establish that NT ancestry information is *present* in the model's hidden states. Our extension asks a sharper question: is this information also *used* by the model when generating subsequent tokens? We investigate this via activation patching—surgically replacing the hidden state at specific positions and measuring the effect on the model's output.

We focus on level NT5 as the target. NT5 sits close enough to the terminal layer that corrupting a single NT5 constituent affects a meaningful but bounded portion of the output string: a typical NT5 subtree spans a few tokens, so replacing its hidden-state summary visibly disrupts the local output without changing the global structure of the entire sequence. Choosing a higher level (e.g. NT2) would corrupt so much of the parse tree that the entire output would collapse; choosing NT6 (one step above terminals) would affect only one or two tokens, making any signal hard to detect.

5.2 Finding the Encoding Layer

Before patching, we need to identify the earliest layer at which NT5 information is fully encoded. We train a diagonal linear probe per layer (3,000 iterations each, evaluated on 500 fresh strings) and record boundary-position accuracy at each layer. Results are shown in Table 4.

NT5 information first reaches high accuracy at layer 5 (98.6%), with all subsequent layers above 98%. We therefore select **layer 5** as the intervention point: it

Table 4: Diagonal probe accuracy at NT5 boundary positions, per layer.

Layer	NT5 accuracy
0	38.1%
1	26.3%
2	51.1%
3	85.2%
4	75.0%
5	**98.6%**
6	98.3%
7	99.7%
8	99.9%
9	99.8%
10	99.5%
11	99.7%

is the earliest layer where the NT5 identity is reliably encoded, meaning any downstream layers that use this information must read it from here.

5.3 Activation Patching Experiment

At inference time, we run the model on a *base* string and replace the hidden states at NT5 boundary positions at layer 5 with states from one of three alternative sources:

Donor. Hidden states from a different string whose NT5 constituent has a different identity. This tests whether swapping the NT5 label at the boundary causally alters the output structure.

Noise (boundary). Gaussian noise injected at NT5 boundary positions only.

Noise (non-boundary). Gaussian noise injected at non-boundary positions only. This is the control: if structure is stored at boundaries, corrupting non-boundaries should have little effect.

We measure CYK validity of the generated output and n-gram KL drift relative to the clean baseline across 500 trials. Results are shown in Table 5.

Table 5: Activation patching results at NT5, layer 5 (N=500 trials).

Condition	CYK valid	CYK drop
Clean	98.8%	—
Donor	41.2%	57.6%
Noise (boundary)	69.8%	29.0%
Noise (non-boundary)	94.0%	4.8%

5.4 Discussion

The donor condition produces a 57.6% drop in CYK validity, confirming that the NT5 identity stored at boundary positions is causally used by the model: replacing it with a structurally incompatible state causes the model to generate invalid strings at a high rate. The noise-at-boundary condition also causes a significant drop (29.0%), further supporting that boundary hidden states carry load-bearing structural information.

Crucially, noise injected at *non-boundary* positions causes almost no disruption (4.8% drop), closely matching the clean baseline of 98.8%. This asymmetry between boundary and non-boundary interventions is a strong causal confirmation of the original paper's hypothesis: the model does not distribute structural information uniformly across all positions, but concentrates it at constituent boundaries, from which downstream layers read it to guide generation.

6 Discussion

Our results confirm the central qualitative findings of Allen-Zhu and Li [1]: after training on next-token prediction over CFG-generated strings, a GPT-2 model does develop internal representations that are sensitive to parse-tree structure. The attention analysis (R6–R9) is particularly striking—the DP-like attention pattern at constituent boundaries emerges clearly even at 6,500 training steps.

The quantitative gaps in completion accuracy and KL divergence are expected consequences of our reduced training budget. The paper's fully-trained models (100,000 steps) achieve near-zero KL divergence on `cfg3b`, whereas our model at 6,500 steps is still converging. The greedy decoding collapse at zero prefix is a further sign of incomplete training: a fully-trained model would assign most probability mass to valid continuations, making greedy decoding more stable.

One notable observation is that greedy decoding with a 50-token prefix (16.50%) outperforms multinomial sampling with the same prefix (8.50%), which is unusual. We attribute this to the prefix providing enough context to suppress the model's residual uncertainty, making greedy selection more reliable in that regime.

7 Conclusion

We have reproduced the experimental setup of Allen-Zhu and Li [1] on the `cfg3b` grammar using a GPT-2 model trained for 6,500 steps. Our results confirm the paper's qualitative findings—particularly the DP-like attention pattern at constituent boundaries and the sensitivity of linear probes to parse-tree structure—while quantitative gaps in KL divergence and completion accuracy reflect our reduced training budget. The codebase is available at https://github.com/annapatata/llm_physics.

Acknowledgements

This work was completed as a course project in Speech and Language Processing at the National Technical University of Athens.

References

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